Code No: 9...

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (ECE & CSE) III - Semester (MAIN)
Examinations Dec-2024/Jan 2025

ENGINEERING MATHEMATICS – III

(PDE & Numerical Methods)

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** is compulsory and answer any **four questions** from the remaining questions (2- 7), each Question carries **12** Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any, may suitably be assumed.***

		Marks	BT	CO
1.a)	Obtain a partial differential equation by eliminating the arbitrary function from $x+y+z=f(x^2+y^2+z^2)$	2	2	1
b)	Classify the following partial differential equation $(1-y)\frac{\partial^2 u}{\partial x^2}+2x\frac{\partial^2 u}{\partial x\partial y}+(1+y)\frac{\partial^2 u}{\partial y^2}=0.$	2	4	2
c)	Find by Newton-Raphson method correct to four decimal places the root between 0 and 1 of the equation $3x-\cos(x)-1=0$	2	5	3
d)	Find the value of $y(x)$ at $x=1.05$ from the table given below :	2	4	4
e)	Evaluate $\int_0^{10}\frac{dx}{1+x^2}$, by using Simpson's $-\frac{1}{3}$ rule, with width $h=1$.	2	1	5
f)	Using Taylor's method, find $y(0.1)$ correct to three decimal places from $\frac{dy}{dx}+2xy=1, y(0)=0$, with $h=0.1$	2	3	5

x	1	1.1	1.2	1.3	1.4	1.5
$y(x)$	0.841	0.891	0.932	0.964	0.985	1.015

2.a)	Solve : $(D^2+3DD'+2D'^2)z=x+y$	6	5	1
b)	Find a general solution of the differential equation $(4D^2+3DD'-D'^2-D-D')z=3e^{(x+2y)/2}$	6	2	1
3.a)	Solve the PDE by separation of variables method : $u_{xx}=u_y+2u$ *where* $u(0,y)=0, \frac{\partial}{\partial x}u(0,y)=1+e^{-3y}$	6	1	2

b) A tightly stretched string with fixed end points $x = 0$ *and* $x = l$ is initially at rest in its equilibrium position. It is set vibrating by given each point a velocity x (l-x) show that displacement is

$$y(x, t) = \frac{8\lambda l^2}{a\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \sin\left\{\frac{(2n-1)}{l}\pi x\right\} \sin\left\{\frac{(2n-1)}{l}\pi at\right\}$$

6 4 2

4.a) Use Gauss elimination method to solve the system of equations $2x + y + z = 10; 3x + 2y + 3z = 18; x + 4y + 9z = 16.$

6 6 3

b) A tightly stretched string with fixed end points $x = 0$ *and* $x = l$ is initially in equilibrium position. It is set vibrating by giving to each of its points an initial velocity given by $\frac{\partial u}{\partial t}(x,0) = u_0 \sin^3(\frac{\pi x}{l})$. Find the displacement $u(x,t)$.

6 2 3

5.a) Explain LU decomposition method.

6 3 4

b) The velocity v $[m/s]$ of a particle at distance s $[m]$ from a point on its path is given by the table below:

s	0	10	20	30	40	50	60
v	47	58	64	65	61	52	38

Estimate the time taken to travel 60 meters by using Simpson's $\frac{1}{3}$ rule.

6 2 4

6.a) Using Runge-Kutta method of fourth order, find $y(0.1), y(0.2)$ and $y(0.3)$, given that

$$\frac{dy}{dx} = 1 + xy;\ y(0) = 2$$

6 4 5

b) Use Lagrange's interpolation formula to fit a polynomial for the following data and find the value of $y(x)$ when $x = 2$

6 6 5

x	0	1	3	4
$y(x)$	−12	0	6	12

7.a) Find the real root of $x^3 - 3x + 1 = 0$, lying between 1 and 2 upto three decimal places by Newton-Raphson method.

6 2 4

b) Find a real root of the equation $x^3 - x - 11 = 0$ by using bisection method and correct to three decimal places.

6 1 5

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